ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. If the categories are mixed together, the domain-specific terms of the category we are interested in may not be selected. When the dataset is very large, mining it as a whole also wastes time and computational cost.

AI used for this answer: Yes

If yes, how: translation

External references (links): https://en.wikipedia.org/wiki/Association_rule_learning

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Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

1. If 40% of the records are dropped because the category_name label is missing, we may destroy the other parts of those records that are still complete and valuable for analysis. In particular, if certain kinds of data are concentrated among the records with missing labels, that valuable part of the data would be lost.
2. One alternative is to group the text documents using clustering. Because clustering does not need labels, the 40% of the data can still be used.

AI used for this answer: Yes

If yes, how: translation

External references (links):

https://pandas.pydata.org/docs/user_guide/missing_data.html

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

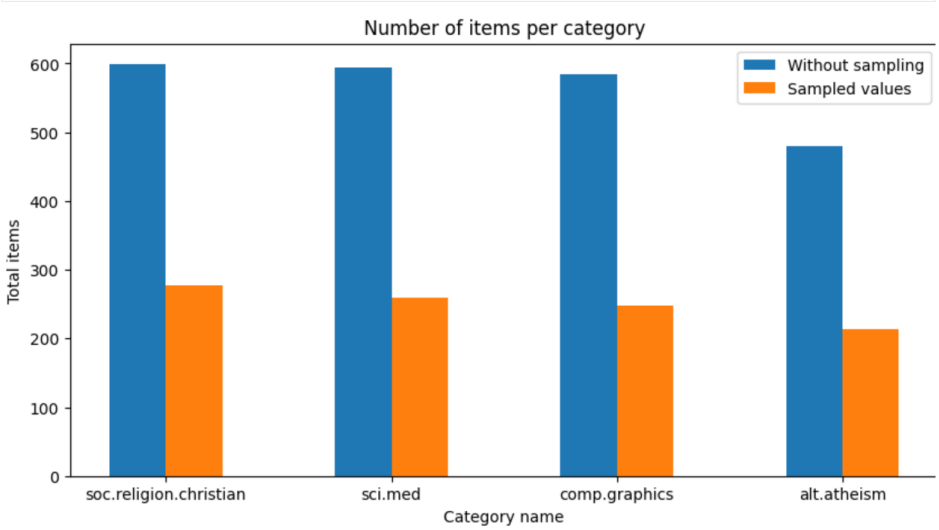
Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, not perfectly, but the distribution of the categories remains roughly the same.



2. With simple random sampling, categories that make up a very small share of the data may not be selected at all.

AI used for this answer: Yes

If yes, how: translation

External references (links): https://en.wikipedia.org/wiki/Stratified_sampling

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The number of features increases: the columns go from 35,482 to 35,788.
2. The sparsity decreases because stop words appear in many documents, so there are fewer zeros in the matrix.

AI used for this answer: Yes

If yes, how: grammar polishing

External references (links):

https://scikit-learn.org/stable/modules/feature_extraction.html

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The total size of the vocabulary decreases as the variance threshold increases from 0.0 to 0.05.
2. Terms that appear in almost no documents, such as nonsense words and meaningless terms, tend to have low variance and are eliminated first.
3. Because these features provide little useful information, removing them usually does not affect model accuracy.

35477	zzz	0.000443	False
35478	kuis	0.000443	False
35479	c5kxix	0.000443	False
35480	c5l184	0.000443	False
35481	000406	0.000443	False

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: grammar polishing

External references (links):

https://scikit-learn.org/stable/modules/feature_selection.html

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Although the English stop words were removed, the top Pearson words still include many everyday words (e.g., "thanks", "hi"), whereas the top Spearman words, which are computed on ranks, are more related to graphics (e.g., "graphics", "image").
2. Pearson correlation measures the strength of a linear relationship, while Spearman correlation measures the relationship between the ranks of the values.
3. A word would have a high Spearman correlation but a much lower Pearson correlation when it appears only once or twice in most of the relevant documents but dozens of times in a few of them.

AI used for this answer: Yes

If yes, how: grammar polishing, concept explanation

External references (links):

https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup: When minSup decreases from 9 to 3, the runtime increases from 0.17 seconds to 3.88 seconds, and the number of patterns increases from 710 to 9,547.

Answer (2) — Top-K k vs MaxFPGrowth minSup: k determines how many patterns are kept, while minSup determines the minimum number of times a pattern must occur.

Answer (3) — Maximal vs standard frequent patterns: Maximal frequent patterns are more concise because they remove the smaller frequent patterns that are subsets of larger frequent patterns.

AI used for this answer: Yes

If yes, how: grammar polishing, concept explanation

External references (links): https://github.com/UdayLab/PAMI

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer: Because X_counts has lots of dimensions, but a scatter plot can only show 2 or 3 dimensions. Apply dimensionality reduction algorithms first.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

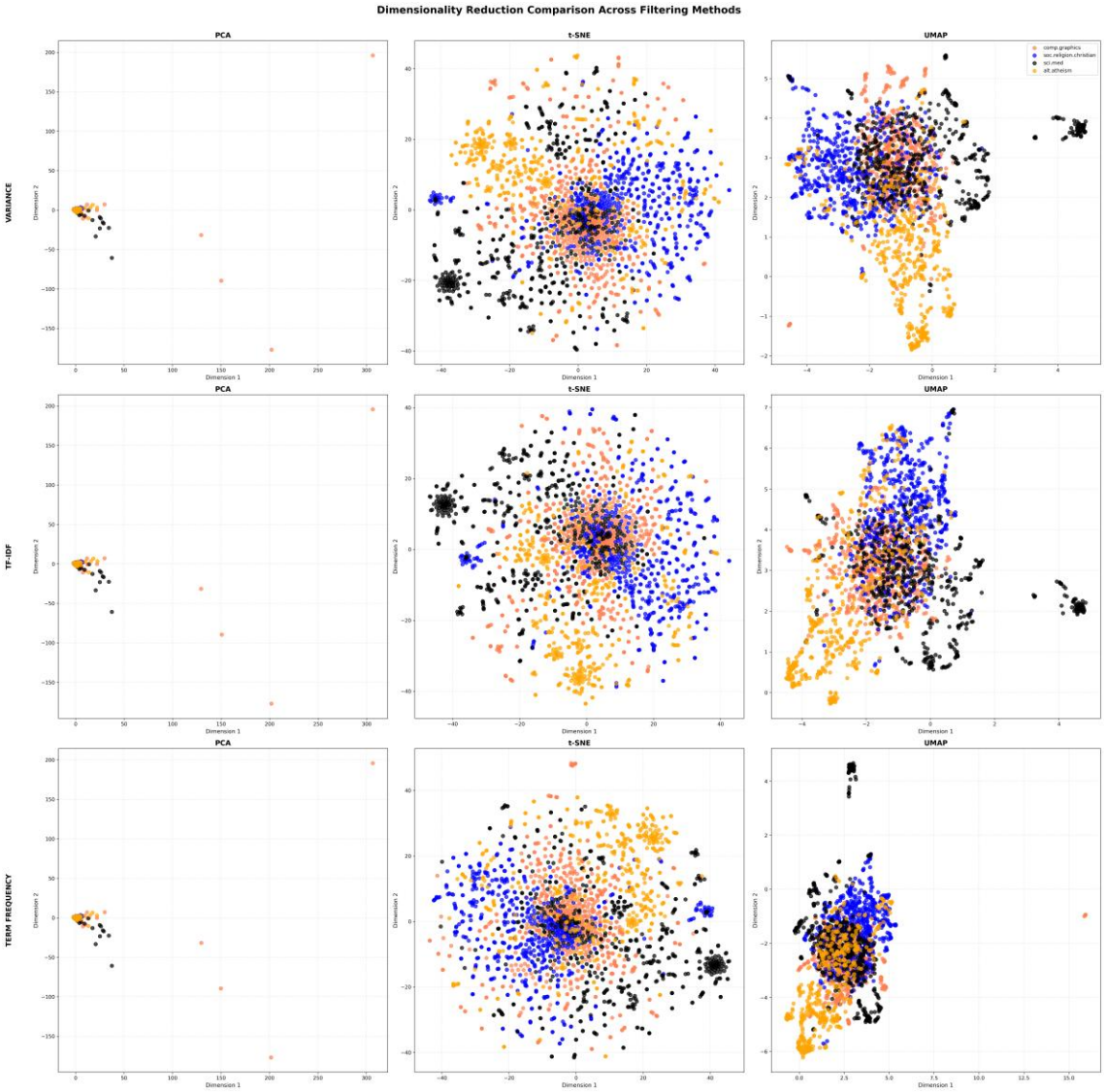
Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. In the PCA plot, almost all the points are crowded together, while t-SNE and UMAP separate the groups more clearly.
2. Because the words in text data rarely have linear relationships with each other, PCA does not reduce the dimensions as well as the non-linear methods t-SNE and UMAP.
3. t-SNE preserves local neighborhood relationships, so the distances after dimensionality reduction do not represent the distances in the original space.



AI used for this answer: Yes

If yes, how: translation, grammar polishing, concept explanation

External references (links): https://scikit-learn.org/stable/modules/manifold.html

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?

3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four binary columns are created.
2. It mistakenly assumes that the categories have a numerical order and that the distances between them are meaningful.
3. Any two categories differ by two bits so the distance between them is the same.

AI used for this answer: Yes

If yes, how: translation

External references (links): https://scikit-learn.org/stable/modules/preprocessing.html

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